

Table 1: Key terminologies commonly used in blockchain technology

Name	Description
Block	A data structure containing a batch of verified transactions.
Hash	A unique alphanumeric string that is derived using a mathematical function to identify blocks and their contents securely.
Consensus algorithm	A protocol used in blockchain networks to achieve the necessary agreement. A consensus algorithm that requires a participator to solve a numerical problem is complex to solve but easy to verify.
Proof of Work (PoW)	PoW is used in cryptocurrencies, such as Bitcoin, to validate transactions and mine new blocks.
Proof of Stake (PoS)	A consensus mechanism used in blockchain networks to achieve an agreement on the state of the ledger.
Smart contract	A self-executing digital contract encoded with predefined rules and conditions, stored and executed on a blockchain platform.
Mining	A process by which new transactions are validated and added to the blockchain, thereby creating new blocks.
Distributed ledger	A decentralised database that maintains a continuously growing list of records or blocks across multiple networked nodes or computers.

are interconnected through the concept of hashing.

Step 6: These authorised transactions are recorded in a public ledger. Once a block is incorporated into an existing chain, transactions are finalised, and the ledger is updated.

Hash function working in a blockchain network

Hashing, a cryptographic technique, generates a fixed-length output when data is processed through a function. The SHA-256 hashing function is employed in the Bitcoin blockchain network. The primary objective of hash functions is to prevent the discovery of any two inputs that yield identical outputs. Two critical characteristics of a hash function are: (i) a minor alteration in a single character completely alters the hash value, and (ii) it operates as a one-way function, which means it cannot be reversed. Figure 2 demonstrates the SHA-256 hash function within the blockchain network.

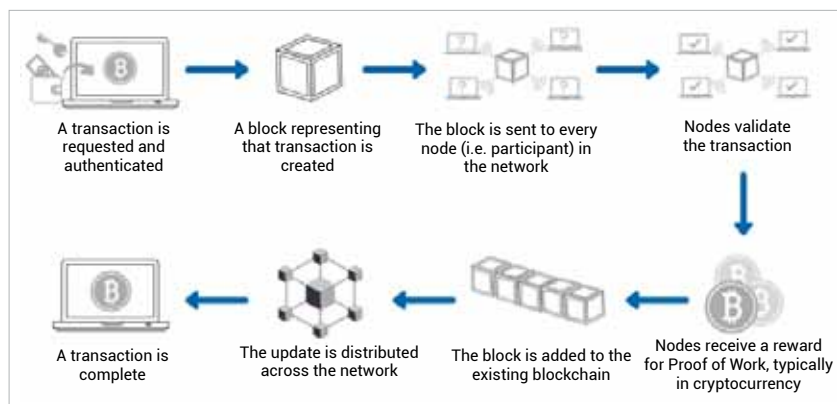


Figure 1: How a block gets added to the blockchain network



Figure 2: Hash function in blockchain